

OSSP-SKILL-WEEK3

Session 5

Task1: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Output:

```
sathwika@123:~$ nano session5_task1.c
sathwika@123:~$ gcc session5_task1.c -o session5_task1
sathwika@123:~$ ./session5_task1
Session 5 - Single Quote Parsing
Enter a command with single quotes:
myshell> echo 'hello $USER'

Quoted strings found:
Quoted String 1: [hello $USER]

Parsing validation:
Single-quote parsing successful.
sathwika@123:~$
```

Observation: I learned how to process single-quoted strings and preserve their contents as literal text. I understood that variable expansion should not occur inside single quotes. I also learned how to store quoted strings, validate the parsing result, and handle errors such as an unclosed single quote.

Task2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Output:

```
sathwika@123:~$ nano session5_task2.c
sathwika@123:~$ gcc session5_task2.c -o session5_task2
sathwika@123:~$ ./session5_task2
Session 5 - Double Quote Parsing
Double quotes preserve spaces and allow variable expansion.
Enter a command: echo "Hello $USER"

Tokens before variable expansion:
Token 1: [echo]
Token 2: [Hello $USER]

Variable expansion:
Token 1: [echo]
Token 2: [Hello sathwika]

Parsing validation:
Double-quote parsing successful.
Spaces inside quotes were preserved.
Variable expansion was processed.
sathwika@123:~$
```

Observation: I learned how to process double-quoted strings while preserving spaces within them. I understood how variable expansion can be performed inside double quotes, such as expanding

environment variables like \$USER. I also learned how to parse quoted and nested tokens, validate the parsing results, and detect errors such as unclosed double quotes.

Session 6

Task1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

Output:

```
sathwika@123:~$ nano session6_task1.c
sathwika@123:~$ gcc session6_task1.c -o session6_task1
sathwika@123:~$ ./session6_task1
Session 6 - Escape Sequence Parsing
Enter a command with escaped spaces or special characters:
myshell> hello\ world

Original input:
hello\ world

Processed input:
hello world

Parser validation:
Escape sequence parsing successful.
Escaped characters were preserved.
sathwika@123:~$
```

Observation: I learned how to handle escape sequences while processing command input. I understood how escaped spaces and special symbols can be preserved as literal characters instead of being treated as separators or operators. I also learned how to detect escape characters, process escaped input, validate parser output, and handle complex user inputs.

Task2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

Output:

```
sathwika@123:~$ nano session6_task2.c
sathwika@123:~$ gcc session6_task2.c -o session6_task2
sathwika@123:~$ ./session6_task2
Session 6 - Process Execution

Parent Process
Parent PID: 823
Child PID: 824
Parent waiting for child...
Child Process
Child PID: 824
Parent PID: 823
Executing ls -l...
total 456
-rw-r--r-- 1 sathwika sathwika 109 Aug 7 05:17 File1.txt
-rw-r--r-- 1 sathwika sathwika 109 Aug 7 05:19 File2.txt
-rw-r--r-- 1 sathwika sathwika  0 Aug 5 03:41 LICENSE
-rw-r--r-- 1 sathwika sathwika  0 Aug 5 03:41 Makefile
```

```

drwxr-xr-x 4 sathwika sathwika 4096 Aug 11 06:30 OSSP
-rw-r--r-- 1 sathwika sathwika 0 Aug 5 03:41 README.md
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 assets
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 bin
-rwxr-xr-x 1 sathwika sathwika 16528 Aug 11 06:59 command_executor
-rw-r--r-- 1 sathwika sathwika 1387 Aug 11 06:59 command_executor.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 docs
-rwxr-xr-x 1 sathwika sathwika 16216 Aug 7 05:19 file_copy
-rw-r--r-- 1 sathwika sathwika 1113 Aug 7 05:17 file_copy.c
-rwxr-xr-x 1 sathwika sathwika 16432 Aug 13 08:28 fork_exec
-rw-r--r-- 1 sathwika sathwika 947 Aug 13 08:28 fork_exec.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 include
-rwxr-xr-x 1 sathwika sathwika 16192 Aug 7 05:39 keyboard_input
-rw-r--r-- 1 sathwika sathwika 1803 Aug 7 05:39 keyboard_input.c
-rwxr-xr-x 1 sathwika sathwika 16352 Aug 7 05:09 main_loop
-rw-r--r-- 1 sathwika sathwika 800 Aug 7 05:09 main_loop.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 obj
drwxr-xr-x 3 sathwika sathwika 4096 Aug 13 08:37 ossp_2520090210
-rwxr-xr-x 1 sathwika sathwika 16824 Aug 11 06:41 parser_double_quotes
-rw-r--r-- 1 sathwika sathwika 3934 Aug 11 06:41 parser_double_quotes.c
-rwxr-xr-x 1 sathwika sathwika 16560 Aug 11 06:34 parser_quotes
-rw-r--r-- 1 sathwika sathwika 2249 Aug 11 06:34 parser_quotes.c
-rwxr-xr-x 1 sathwika sathwika 15952 Aug 1 04:49 process
-rw-r--r-- 1 sathwika sathwika 80 Aug 1 04:48 process.c
-rw----- 1 sathwika sathwika 80 Aug 1 04:46 process.c.save
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 scripts
-rwxr-xr-x 1 sathwika sathwika 16224 Aug 13 12:04 session2_task1
-rw-r--r-- 1 sathwika sathwika 559 Aug 13 12:04 session2_task1.c
-rwxr-xr-x 1 sathwika sathwika 16096 Aug 13 12:10 session2_task2
-rw-r--r-- 1 sathwika sathwika 537 Aug 13 12:10 session2_task2.c
-rwxr-xr-x 1 sathwika sathwika 16632 Aug 15 12:22 session3_task1
-rw-r--r-- 1 sathwika sathwika 4035 Aug 15 12:22 session3_task1.c
-rwxr-xr-x 1 sathwika sathwika 16392 Aug 15 12:26 session3_task2
-rw-r--r-- 1 sathwika sathwika 2160 Aug 15 12:26 session3_task2.c
-rwxr-xr-x 1 sathwika sathwika 16312 Aug 15 12:34 session4_task1
-rw-r--r-- 1 sathwika sathwika 1279 Aug 15 12:34 session4_task1.c
-rwxr-xr-x 1 sathwika sathwika 16272 Aug 15 12:37 session4_task2
-rw-r--r-- 1 sathwika sathwika 2324 Aug 15 12:37 session4_task2.c
-rwxr-xr-x 1 sathwika sathwika 16184 Aug 15 12:43 session5_task1
-rw-r--r-- 1 sathwika sathwika 1603 Aug 15 12:42 session5_task1.c
-rwxr-xr-x 1 sathwika sathwika 16312 Aug 15 12:54 session5_task2
-rw-r--r-- 1 sathwika sathwika 3214 Aug 15 12:53 session5_task2.c
-rwxr-xr-x 1 sathwika sathwika 16184 Aug 15 12:57 session6_task1
-rw-r--r-- 1 sathwika sathwika 1191 Aug 15 12:57 session6_task1.c
-rwxr-xr-x 1 sathwika sathwika 16352 Aug 15 12:59 session6_task2
-rw-r--r-- 1 sathwika sathwika 1005 Aug 15 12:59 session6_task2.c
-rwxr-xr-x 1 sathwika sathwika 16304 Aug 13 10:14 skill_week1_task2
-rw-r--r-- 1 sathwika sathwika 879 Aug 13 10:13 skill_week1_task2.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 src
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 tests
Child exited with status: 0
Parent process completed.

```

Observation: I learned how to create a child process using `fork()` and execute another program using `exec()`. I understood how to pass arguments to the executed program and how the parent process can wait for the child using `waitpid()`. I also learned how to handle execution errors and check the child's exit status using process-management functions.